

Multiple Choice (10 marks)

1. What is the greatest common divisor of $2^{1001} - 1$ and $2^{1012} - 1$?
 - (a) 1
 - (b) 2049
 - (c) 2048
 - (d) 2047
2. If x and y are directly proportional and $x = 3$ when $y = 8$, what is the value of x when $y = 13$?
 - (a) 0.615
 - (b) 4.875
 - (c) 15
 - (d) 34.667
3. Which of the following points is NOT a solution to both the inequalities $y > 9x - 8$ and $y < -x + 8$?
 - (a) $(-2, 15)$
 - (b) $(-1, 5)$
 - (c) $(0, 0)$
 - (d) $(1, 5)$
4. A hypothesis test is set up so that $P(\text{rejecting } H_0 \text{ when } H_0 \text{ is true}) = 0.05$ and $P(\text{failing to reject } H_0 \text{ when } H_0 \text{ is false}) = 0.26$. What is the power of the test?
 - (a) 0.26
 - (b) 0.05
 - (c) 0.95
 - (d) 0.74
5. Find the value of the expression $-16 + 13 + (-33)$.
 - (a) 4
 - (b) 62
 - (c) 36
 - (d) -36

True / False (10 marks)

Answer True or False

6. True or False: The decimal representation of 7 over 33 is 0.22 repeating.
7. True or False: The result of dividing $-18/7$ by $-6/5$ is $7/15$.
8. True or False: The numbers 5 over 6, 1 over 3, 0.75, and 1 over 8 are in order from least to greatest.
9. True or False: The square of 15 is 225.
10. True or False: The probability that the two numbers rolled on two fair 6-sided dice sum to 4 is $\frac{1}{4}$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. George has a budget of 23 *for art supplies, including markers, paper, and glue. If the combined cost of the* formulate an inequality for the amount, p , he can spend on glue and explain your reasoning.
12. Abby drives 18 miles on her daily round trip to work. Calculate the total miles driven in 5 days and discuss how rounding affects your estimate. Provide a detailed explanation of your reasoning.

End of Paper